

Minas Aslanyan

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Google Scholar | linkedin.com/in/maslanyan

RESEARCH PROFILE

Computer scientist working at the intersection of **human motion modelling, learning-based perception, and real-time interactive systems**. My doctoral research builds closed-loop environments that estimate human movement from vision and inertial sensing, quantify deviation from a reference motion, and return corrective feedback to the user in real time — effectively a perception-to-feedback control loop with a human in the loop. This work spans mobile 3D pose estimation, IMU/vision sensor fusion, skeleton-based motion representation learning, and VR-based training environments. I am seeking to extend it toward **human-centred and data-driven control for physical human-robot interaction**, in particular models of human motor behaviour that a controller can rely on for assistance, rehabilitation, and shared-autonomy tasks.

Interests: human-centred control · human-robot interaction · data-driven modelling of human motion · state estimation and sensor fusion · machine learning for perception · haptic and visual assistance · rehabilitation and motor-learning technology

EDUCATION

Ph.D. in Computer Science

2020 – 2026

Institute for Informatics and Automation Problems (IIAP), National Academy of Sciences of Armenia, Yerevan

- Dissertation: *AI-Driven Environment for Automated Physical and Biomechanical Analysis*.
- Advisor: Dr. Levon Aslanyan, Doctor of Physical and Mathematical Sciences, Head of the Discrete Mathematics Department.
- Topics: real-time 3D human pose reconstruction on mobile hardware, biomechanical error quantification, corrective-feedback generation, and integration with VR training environments.

M.Sc. in Digital Signal Processing

2015 – 2017

Institute for Informatics and Automation Problems (IIAP), National Academy of Sciences of Armenia, Yerevan

- Thesis: *Creation of an Algorithmic Application for Signal Analysis and Evaluation*.

B.Sc. in Electrical and Electronics Engineering

2009 – 2013

State Engineering University of Armenia (National Polytechnic University of Armenia), Yerevan

RESEARCH EXPERIENCE

Researcher / Doctoral Researcher

2020 – present

Institute for Informatics and Automation Problems, NAS RA, Yerevan, Armenia

- Designed and validated a real-time 3D human pose estimation pipeline optimised for commodity mobile hardware, sustaining 30 FPS on mid-range devices under on-device compute and latency constraints.
- Conducted a systematic comparative evaluation of inertial (IMU) versus vision-based motion capture, characterising accuracy, drift, and failure modes of each modality and the conditions under which fusing them is justified.
- Developed occlusion-aware, part-level supervised contrastive learning for skeleton-based gait recognition, improving robustness of motion representations under partial body occlusion.
- Built an intelligent VR workout environment coupling a head-mounted display with an external smartphone as the sensing channel, closing a perception-feedback loop across two devices with tight latency budgets.

- Engineered an LLM-driven guidance layer that translates pose-derived kinematic error signals into adaptive, context-aware corrective instructions for the user.
- Full experimental ownership: study design, participant instruction, motion-data collection, signal processing, statistical analysis, and validation against reference measurements.

Founder and Technical Lead — Jamie: AI Fitness Coach (applied research project) 2023 – 2024

Independently funded research prototype deployed at scale

- Deployed the research pipeline as a public application (25,000+ downloads, 4.7-star rating), yielding a large-scale, in-the-wild evaluation of mobile pose estimation and feedback quality outside laboratory conditions.
- Implemented real-time on-device computer vision (pose estimation, repetition counting) and 3D virtual-trainer agents in Unity/C#, integrated with a Node.js and Python backend.

PUBLICATIONS

Peer-reviewed; sole or first author on all items. Full list: Google Scholar.

Journal Articles

1. M. Aslanyan, “Comparative Evaluation of Inertial Measurement Unit versus Vision-Based Motion Capture,” *Pattern Recognition and Image Analysis*, vol. 35, no. 4, 2025. (Scopus)
2. M. Aslanyan, “On Mobile Pose Estimation and Action Recognition: Design and Implementation,” *Pattern Recognition and Image Analysis*, vol. 34, no. 1, 2024. (Scopus)

Conference Papers

3. M. Aslanyan, “Development of Intelligent Workout Environment for VR Devices,” in *Proc. 2024 IEEE Global Engineering Education Conference (EDUCON)*, Kos Island, Greece, Apr. 2024. ieeexplore.ieee.org/document/10578849
4. M. Aslanyan, “On Mobile Pose Estimation Design and Implementation,” in *Proc. Computer Science and Information Technologies (CSIT)*, Yerevan, Armenia, 2023.

In Preparation / Under Review

5. M. Aslanyan and L. Aslanyan, “Occlusion-Aware Part-Level Supervised Contrastive Learning for Skeleton-Based Gait Recognition.”

TEACHING EXPERIENCE

Adjunct Lecturer 2022 – 2024

Université française en Arménie (French University in Armenia, UFAR), Yerevan

- Lectured Python programming, data structures, and algorithms; designed and graded coursework and supervised undergraduate students through hands-on programming exercises.

Teaching Assistant — Cloud Computing 2023 – 2024

American University of Armenia (AUA), Yerevan

- Supported lecture delivery, graded assignments, and held weekly office hours.

TECHNICAL SKILLS

Programming	Python, MATLAB, C#, Java, C, TypeScript/JavaScript, SQL, Dart
ML & Modelling	PyTorch, deep learning for vision, contrastive representation learning, OpenCV, time-series modelling, LLM orchestration (LangChain), RAG
Signals & Estimation	Digital signal processing, filtering, IMU/vision sensor fusion, motion capture pipelines, real-time low-latency systems
Simulation & HRI	Unity (3D interactive and VR environments, C#), on-device inference, mobile sensing
Tooling	Docker, Git, CI/CD, FastAPI, Node.js, AWS, GCP, PostgreSQL, Linux
Languages	Armenian (native), English (fluent, C1), Russian (fluent), German (beginner, A1 — actively studying)

PROFESSIONAL ENGINEERING EXPERIENCE

Included for completeness; demonstrates large-scale software engineering, team leadership, and delivery of production research systems.

Inferaim — Principal Engineer, SaaS Platform & AI Systems **2024 – 2026**

- Owned architecture of a multi-tenant, event-driven microservice platform on AWS; designed and shipped an LLM-based natural-language-to-SQL engine (retrieval over schema embeddings with an agentic validation loop) reaching 92% accuracy on 10,000+ daily queries.
- Hired, mentored, and led a team of five engineers; established design-review, code-review, and CI/CD standards.

Jamie: AI Fitness Coach — Co-Founder & CTO **2023 – 2024**

- Led a team of four engineers; delivered the mobile motion-analysis system described under Research Experience from prototype to public release.

VMware Inc. — Software Engineer I–III **2014 – 2020**

- Built core visualisation and analytics components of VMware Aria Operations, a large-scale enterprise observability platform (Java, React, D3.js, REST APIs).
- Led a cross-team modernisation of the shared visualisation stack, removing third-party licensing costs of \$3M company-wide; mentored junior engineers and ran internal technical talks.

AWARDS AND RECOGNITION

- **“Top 100 Global Key Talent,”** VMware Inc. — selected from a worldwide engineering organisation, with an accompanying equity award.
- Four projects selected and pitched in VMware Innovation Sprint programmes.

REFERENCES

Dr. Levon Aslanyan — Doctoral advisor

Doctor of Physical and Mathematical Sciences; Head, Discrete Mathematics Department
Institute for Informatics and Automation Problems, NAS RA, 1 Paruyr Sevak St., Yerevan 0014, Armenia
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Further references and full transcripts available on request.